Power Analysis for Research Studies

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Introduction

Key Concepts Overview

- Hypothesis
  - H_0 and H_1
  - Type I error (α)
  - Type II error (β)
- Statistical power
- Effect size
- Sample size (n)

Hypothesis

Definition (C&B.D.8.1.2)

A **hypothesis** is a statement about a population parameter.

- A hypothesis makes a claim about the population
- The goal of hypothesis testing:
  - to use the evidence from sample data
  - to decide which of two complementary hypotheses is more supported

Hypothesis

Definition (C&B.D.8.1.3)

The two complementary hypotheses in a hypothesis testing problem are called **the null hypothesis (H_0)** and **the alternative hypothesis (H_1)**, respectively.

- H_0: No effect or difference, $\mu = \mu_0$
- H_1: There is an effect or difference, $\mu < \mu_0$

Hypothesis Testing

Definition (C&B.D.8.1.3)

A **hypothesis testing procedure** or **hypothesis test** is a rule:
For which sample values the decision is made to accept H_0 as true.
For which sample values H_0 is rejected and H_1 is accepted as true.

- The subset of the sample space for which H_0 will be rejected is called the *rejection region* or *critical region* (R).
- The complement of the rejection region (R^C) is called the *acceptance region*.

Hypothesis Testing Decision Table

Rosner.T.7.1: Four possible outcomes in hypothesis testing

		Decision	
		Accept H_0	Reject H_0
Truth	H_0	H_0 is true H_0 is accepted	H_0 is true H_0 is rejected
	H_1	H_1 is true H_0 is accepted	H_1 is true H_0 is rejected

careful about the header and the labels of the table, make sure they are consistent with the content of the table.

Hypothesis Testing Decision Table

C&B.T.8.3.1: Two types of errors in hypothesis testing

		Decision	
		Accept H_0	Reject H_0
Truth	H_0	Correct decision $(1-\alpha)$	Type I Error (False Positive, α)
	H_1	Type II Error (False Negative, β)	Correct decision (Power $= 1-\beta$)

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False Positive: Type I Error α

False Negative: Type I I Error β

Defintions (Rosner.D.7.2 - D.7.6)

- The probability of a **Type I error** is the probability of rejecting the null hypothesis when H_0 is true.
- The probability of a **Type II error** is the probability of accepting the null hypothesis when H_1 is true. This probability is a function of μ as well as other factors.
- The probability of a Type I error is usually **denoted by** α and is commonly referred to as the significance level of a test.
- The probability of a Type II error is usually **denoted by** β.
- The **power** of a test is defined as $1 - \beta$:
  $\Pr(\text{rejecting } H_0 \mid H_1 \text{ true}) = 1 - \text{probability of a Type II error}$

What is power?

Power $(1 - \beta)$

- The ability of a statistical test to detect a relationship or difference
- The probability of correctly rejecting the null hypothesis when the null hypothesis is false (therefore it must be rejected).

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- controversially, people also do *post-hoc* power analysis

Example Rosner.7.2: Obstetrics

Obstetrics Suppose we want to test the hypothesis that mothers with low socio- economic status (SES) deliver babies whose birthweights are lower than "normal." To test this hypothesis, a list is obtained of birthweights from 100 consecutive, full-term, live-born deliveries from the maternity ward of a hospital in a low-SES area. The mean birthweight (x) is found to be 115 oz with a sample standard deviation (s) of 24 oz. Suppose we know from nationwide surveys based on millions of deliveries that the mean birthweight in the United States is 120 oz. Can we actually say the underlying mean birthweight from this hospital is lower than the national average?

Example Rosner.7.2: Obstetrics

Power Function for One-Sided Test $(H_1 : \mu < \mu_0)\}$

To test $H_0 : \mu = \mu_0 \quad \text{vs.} \quad H_1 : \mu < \mu_0$

$$\begin{aligned}
\text{Power} &= \Pr(\text{reject } H_0 \mid H_0 \text{ false}) \\
&= \Pr\left(Z < z_\alpha \mid \mu = \mu_1\right) \\
&= \Pr\left(\frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} < z_\alpha \,\middle|\, \mu = \mu_1\right) \\
&= \Pr\left(\bar{X} < \mu_0 + z_\alpha \frac{\sigma}{\sqrt{n}} \,\middle|\, \mu = \mu_1\right)
\end{aligned}$$

Under H_1, we know:$\bar{X} \sim N\left(\mu_1, \frac{\sigma^2}{n}\right)$

Hence, standardizing:

$$\text{Power} = \Phi\left(\frac{\mu_0 + z_\alpha \frac{\sigma}{\sqrt{n}} - \mu_1}{\sigma/\sqrt{n}}\right) = \Phi\left(z_\alpha + \frac{\mu_0 - \mu_1}{\sigma}\sqrt{n}\right)$$

Problem: Compute the power of the test for the birthweight data in Example 7.2 with an alternative mean of 115 oz and $\alpha = 0.05$, assuming $\sigma = 24$ oz.

$$\mu_0 = 120, \quad \mu_1 = 115, \quad \alpha = 0.05, \quad \sigma = 24, \quad n = 100$$

$$\text{Power} = \Phi\left[z_{0.05} + \frac{(120 - 115)\sqrt{100}}{24}\right]$$

$$= \Phi\left[-1.645 + \frac{5(10)}{24}\right] = \Phi(0.438) = 0.669$$

There is about a 67% chance of detecting a significant difference at $\alpha = 0.05$.

Power is a function depdent on several factors:

1. effect size
2. α level
3. sample size (very important)
4. statistical test type
5. the type of design and data quality

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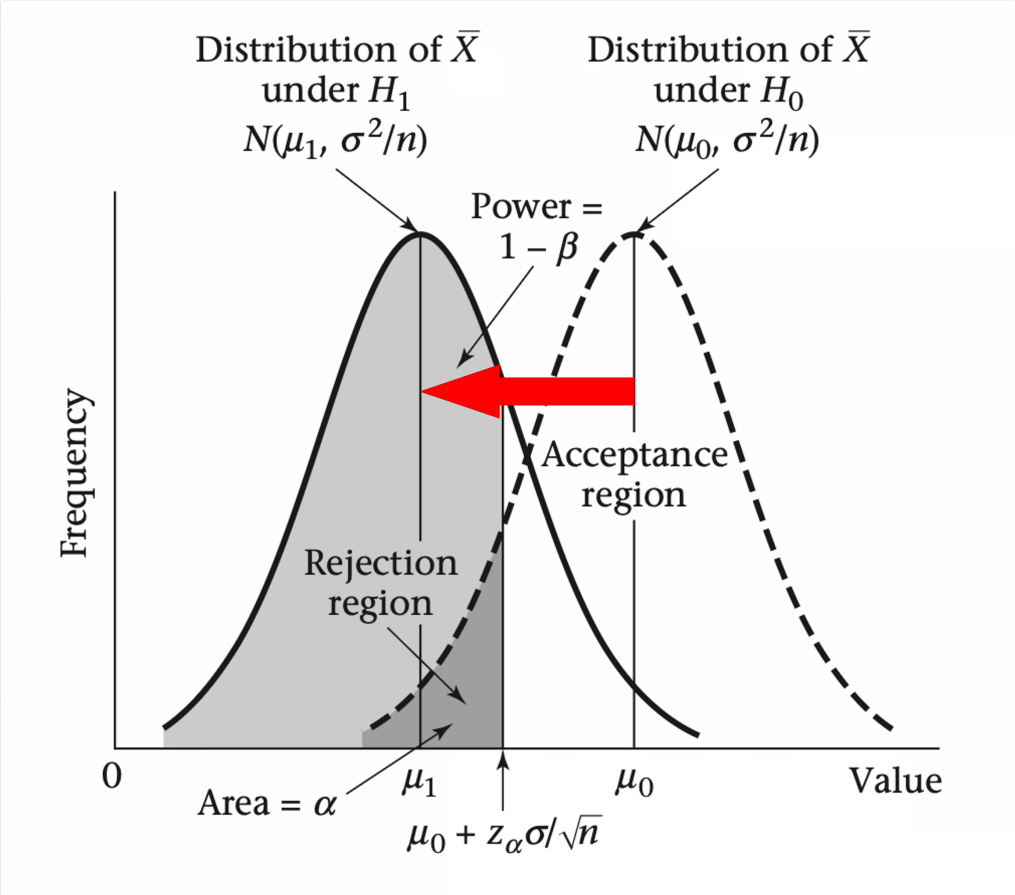
Effect size

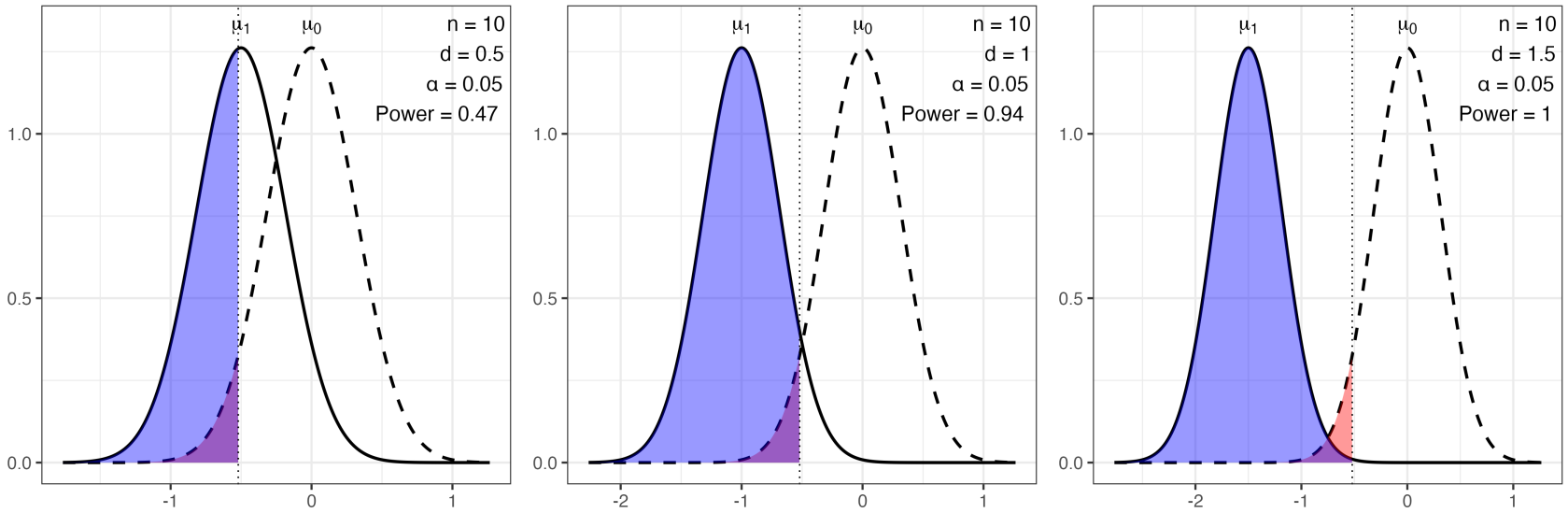
- Effect size is the quantitative measure of the relationship or difference that we want to detect
- Emphasizes the size of the relationship or difference
- Examples:
  - Cohen's d for mean differences for t-test
  - Pearson's r and R^2 for correlations
  - The regression coefficients in a regression (β_0, β_1, etc.)
  - The mean differences in ANOVA (η)



In real life, we cannot change the level of the effect size. The effect size is determined by the nature of the population phenomenon.

Effect size

- If the effect size is small, then we will need more subjects to find a significant results.
- If the effect size is large, then we do not need many subjects to find a significant results (10 or fewer).



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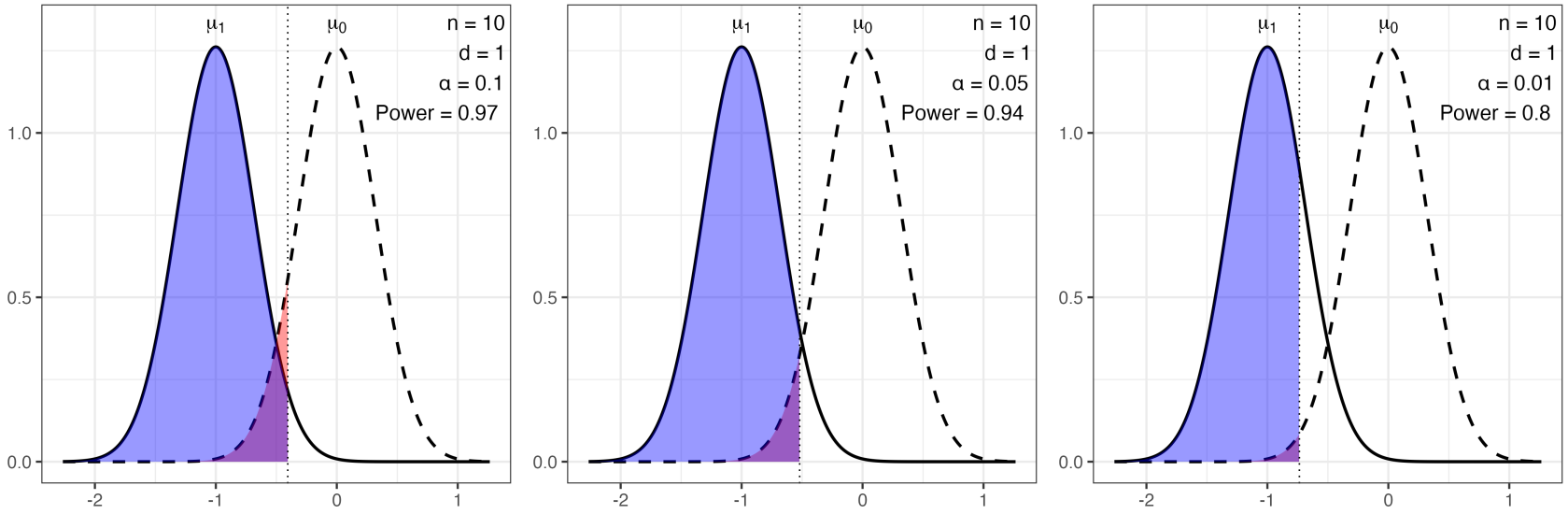
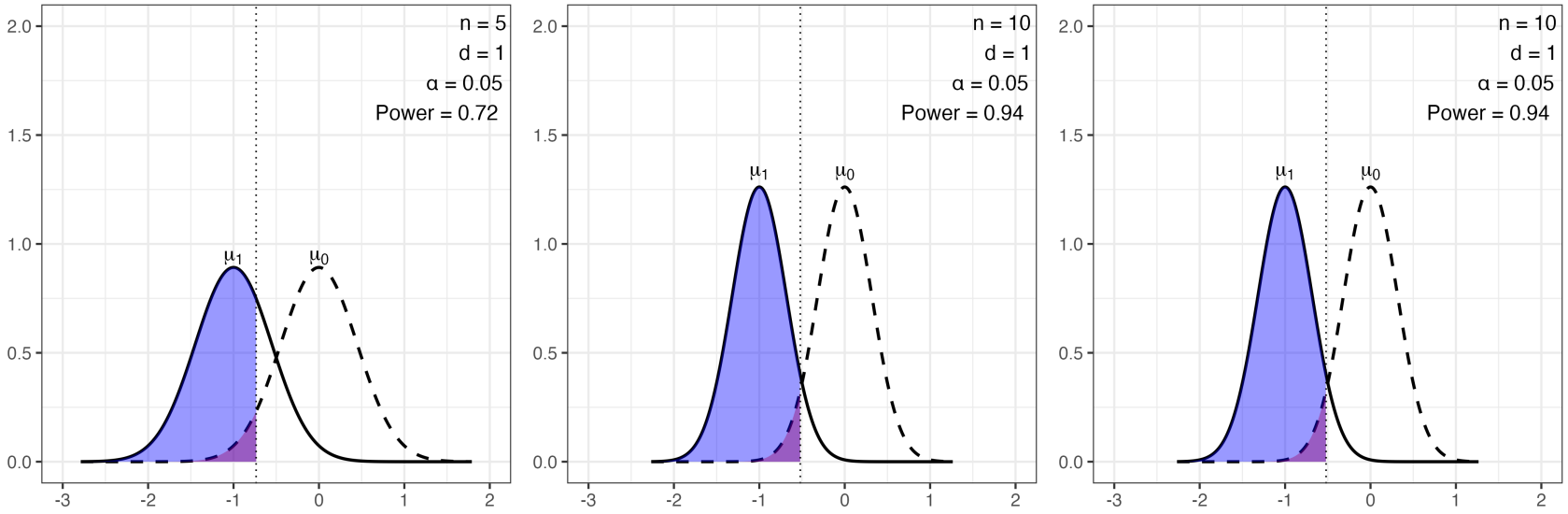

μ1
μ0
n = 10
d = 1
α = 0.1
Power = 0.97
μ1
μ0
n = 10
d = 1
α = 0.05
Power = 0.94
μ1
μ0
n = 10
d = 1
α = 0.01
Power = 0.8
1.0
0.5
0.0
-2
-1
0
1


n = 5
d = 1
α = 0.05
Power = 0.72
n = 10
d = 1
α = 0.05
Power = 0.94
n = 10
d = 1
α = 0.05
Power = 0.94
μ1
μ0
2.0
1.5
1.0
0.5
0.0
-3
-2
-1
0
1
2

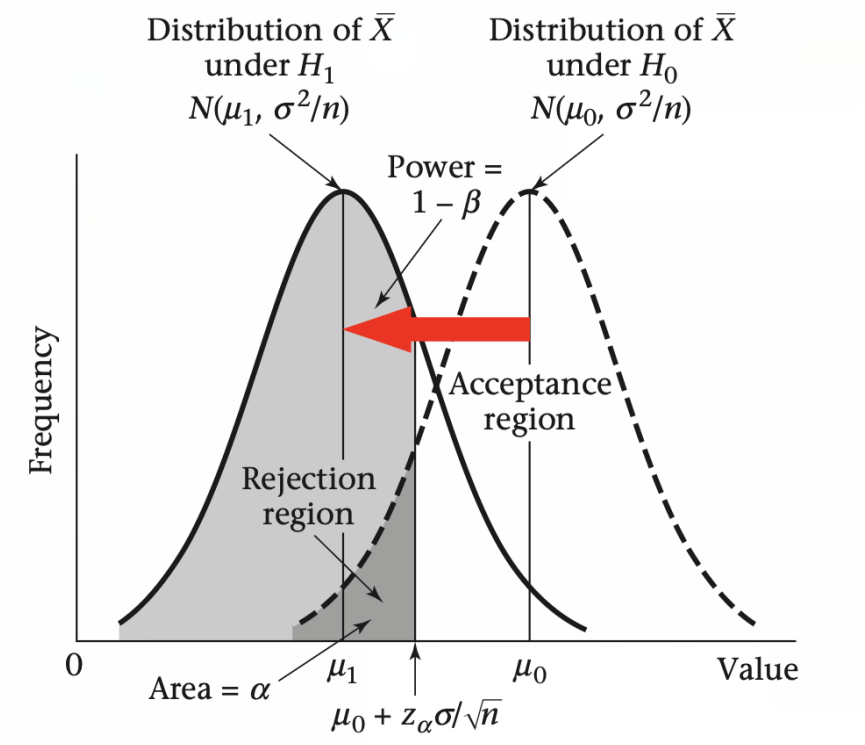
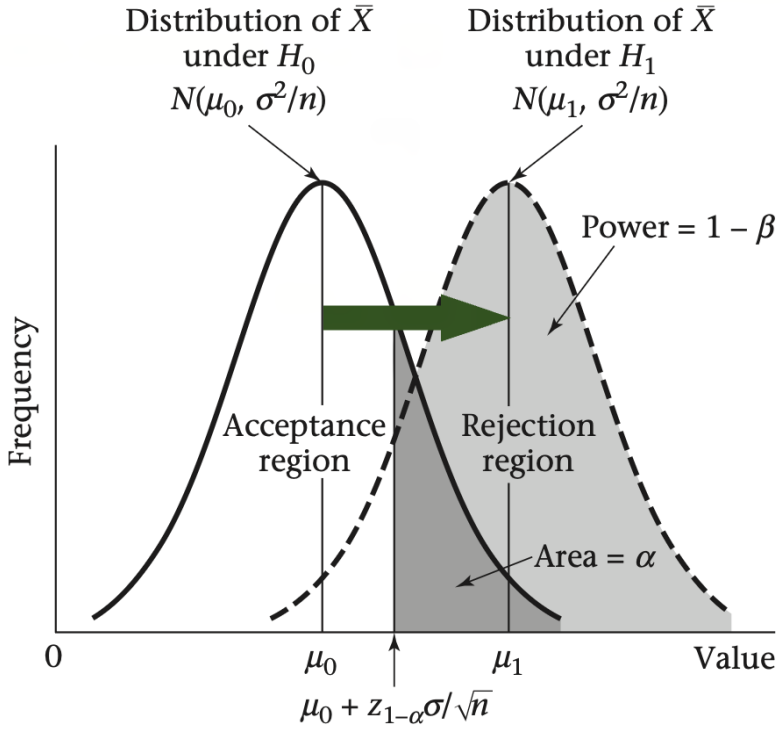
How to improve power (APRONS)

- Relax **A**lpha
- **P**arametric Statstic
- **R**eliability
- **O**ne-tailed test
- increase sample size **N**
- **S**ensitivity of the analysis

Computational Tools

- How to estimate the sample size need to have adequate power
- Where to get the effect size - previous studies
- Alpha level
  - conventionally 0.05, but can be relaxed to larger values
- Software: G*Power, R packages (e.g., pwr)
- Github repositories for power analysis examples and practices!

The last slide

post hoc Type III Error

The **direction** of the effect is opposite to what we expected.